Bivariate Transformations

1. CB 4.15
2. CB 4.17
3. CB 4.19 (For part (a), don't use standard transformation methods. Instead use the results of example 2.1.9 and example 4.3.4. For part (b), the problem should say "find the marginal distributions of $X_1/(X_1 + X_2)$ and $(X_1 + X_2)$")
4. CB 4.20 (you do not need to interpret the result geometrically)
5. CB 4.30 (b only)
6. CB 4.45
7. CB 4.52
8. CB 4.53
9. Let X and Y be independent random variables with the same geometric distribution.
   (a) Find the joint pmf of X and $X + Y$.
   (b) Find the marginal distribution of $X + Y$ using transformation methods (not the mfg method)
10. Let (X, Y) have joint pdf $f(x, y) = x + y$ for $0 \leq x \leq 1$ and $0 \leq y \leq 1$ and 0 otherwise. Let $U = 2X$ and $V = X + Y$. Find the joint pdf of (U, V) and the marginal pdfs of U and V.
11. Suppose the random variables X and Y have joint density function

$$f(x, y) = xy/96 \qquad 0 < x < 4; 1 < y < 5.$$

    Find the pdf of $U = X + 2Y$.
12. A current of X flows through a resistance of Y, thus generating the power $W = X^2Y$. Suppose that X has a pdf $f(x) = 2x$, $0 \leq x \leq 1$, that Y is uniform(0,1), and that X and Y are independent. Find the pdf of W. (*Hint:* First find the joint pdf of W and X.)